

NPTEL

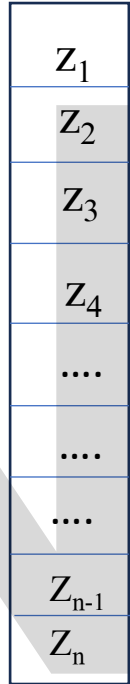
StyleGAN

Contents

- Limitation of Traditional GAN and Motivation for StyleGAN
- Latent Space Entanglement
- Introduction to StyleGAN
- Mapping Network
- Affine Transformation
- Synthesis Network
- Adaptive Instance Normalization
- Style Based Generator

Traditional GAN

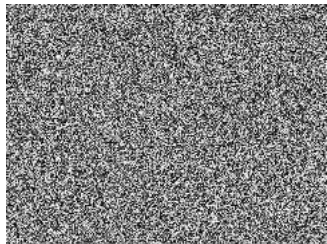
Latent Vector



Generator
(G)



$G(z) \rightarrow \text{image}$

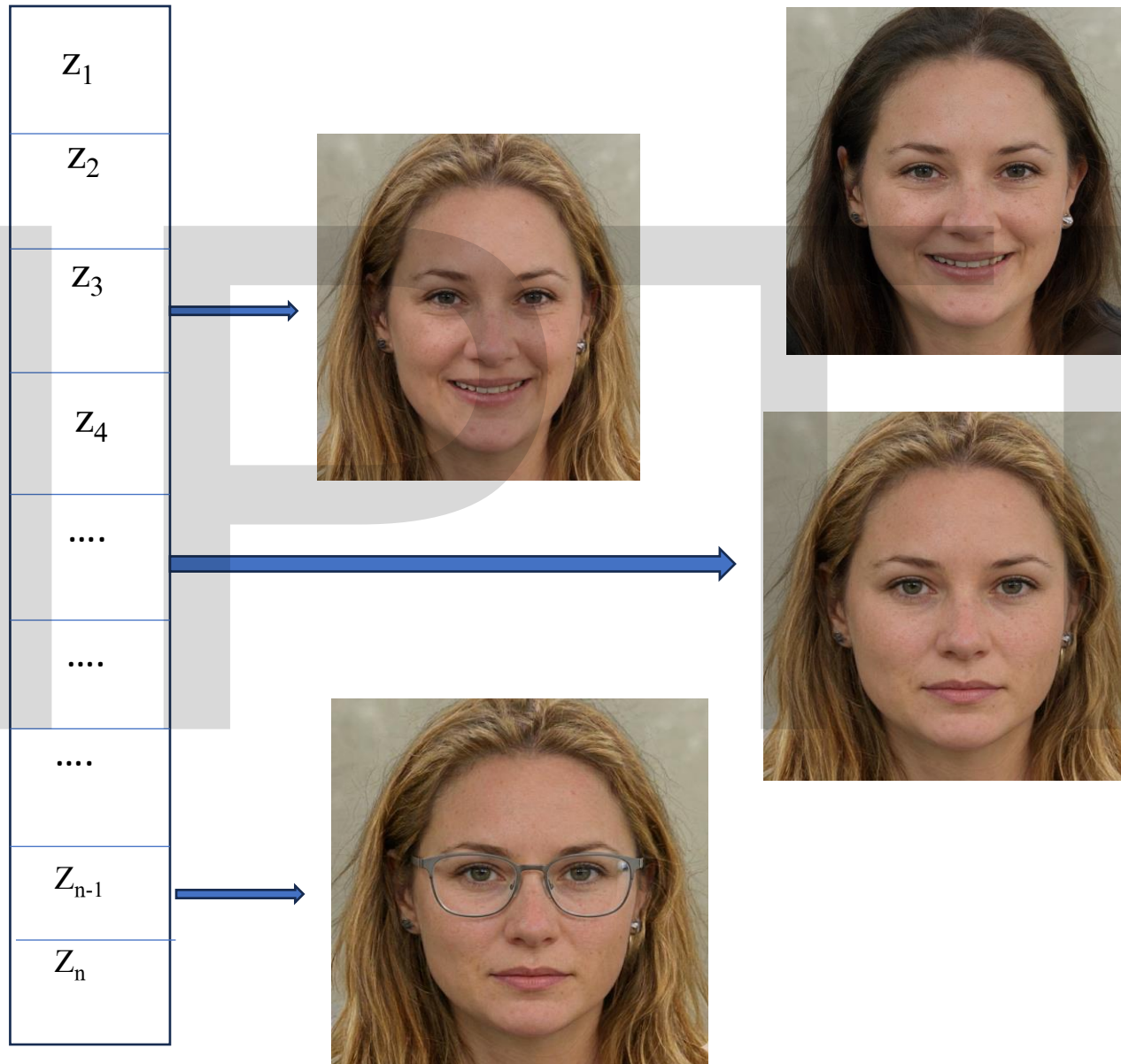


Noise

In a standard Generative Adversarial Network
We sample a random vector:

$$z \sim N(0, 1)$$

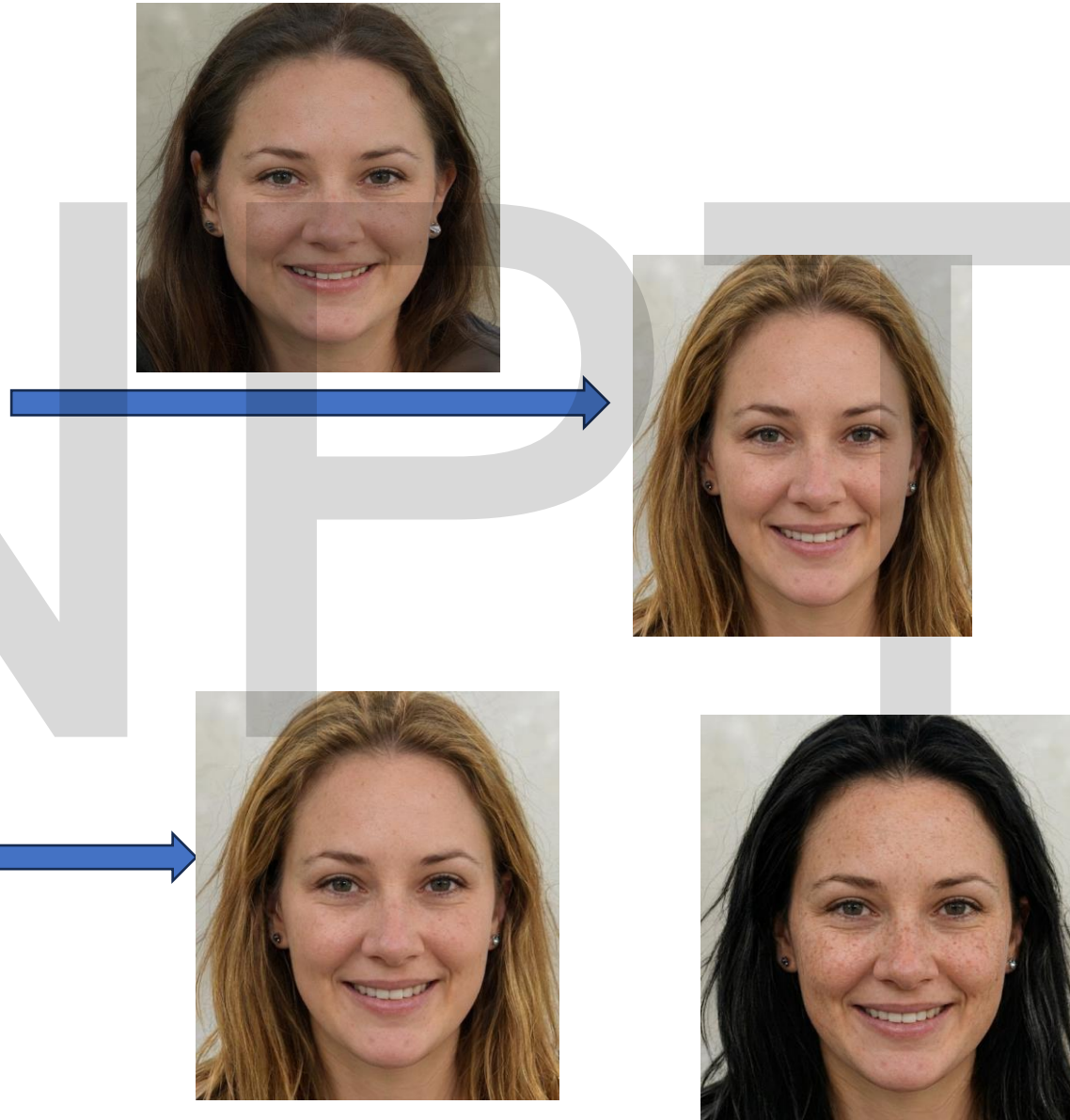
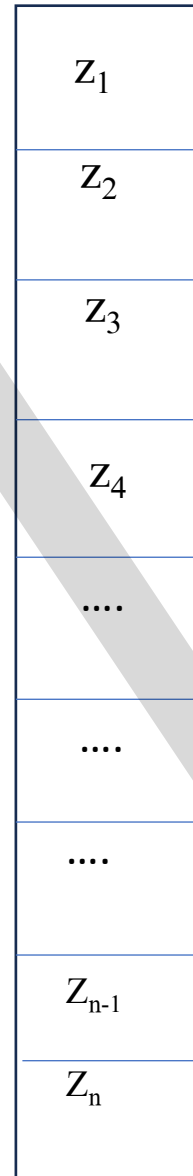
Motivation for StyleGAN



Researchers wanted finer control
over generated image attributes

That was not possible in traditional
GAN

Latent Space Entanglement



✓ One latent variable affects multiple image attributes simultaneously.

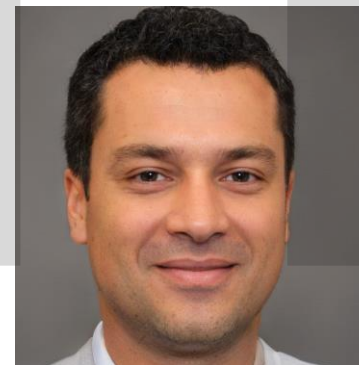
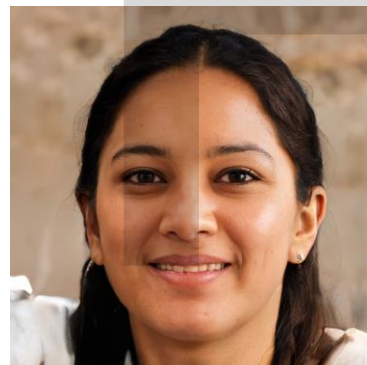
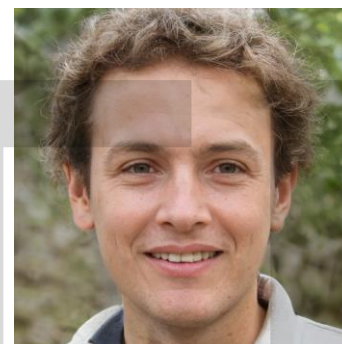
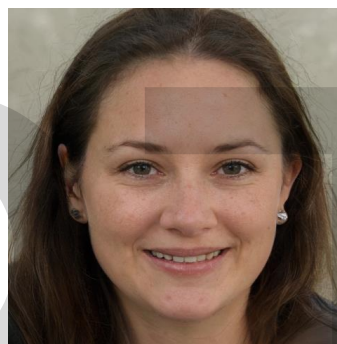
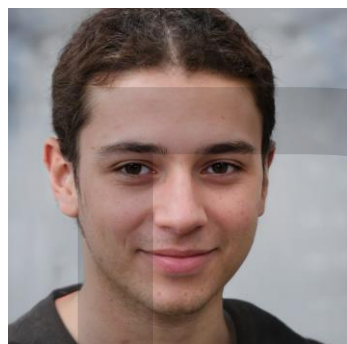
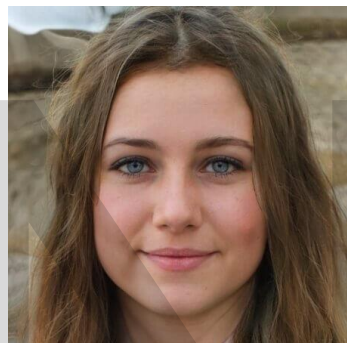
✓ changing one latent dimension affects multiple semantic attributes

WHY?

“The input latent space must follow the probability density of the training data, and this leads to unavoidable entanglement.”

Why latent Space Entanglement Happens

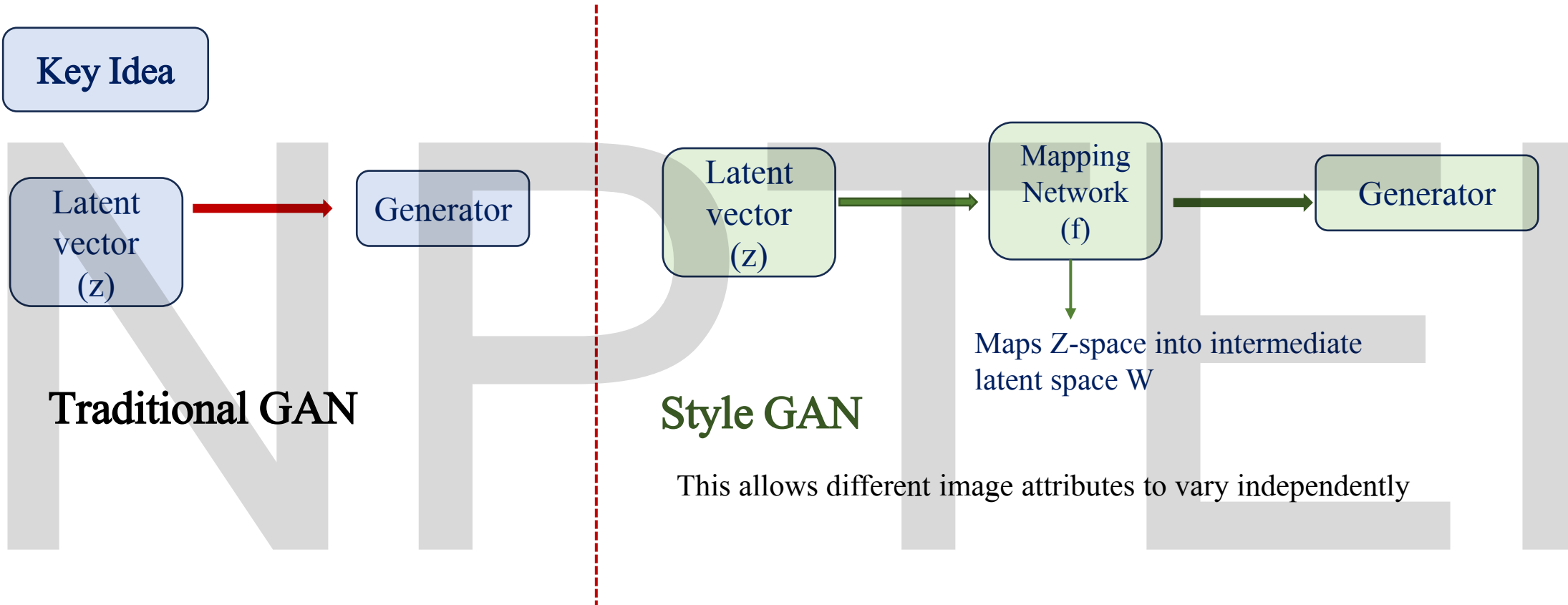
Dataset



- Multiple facial attributes become correlated
- Latent space becomes curved because the generator avoids producing unlikely combinations.

Introduction to StyleGAN

Key Idea

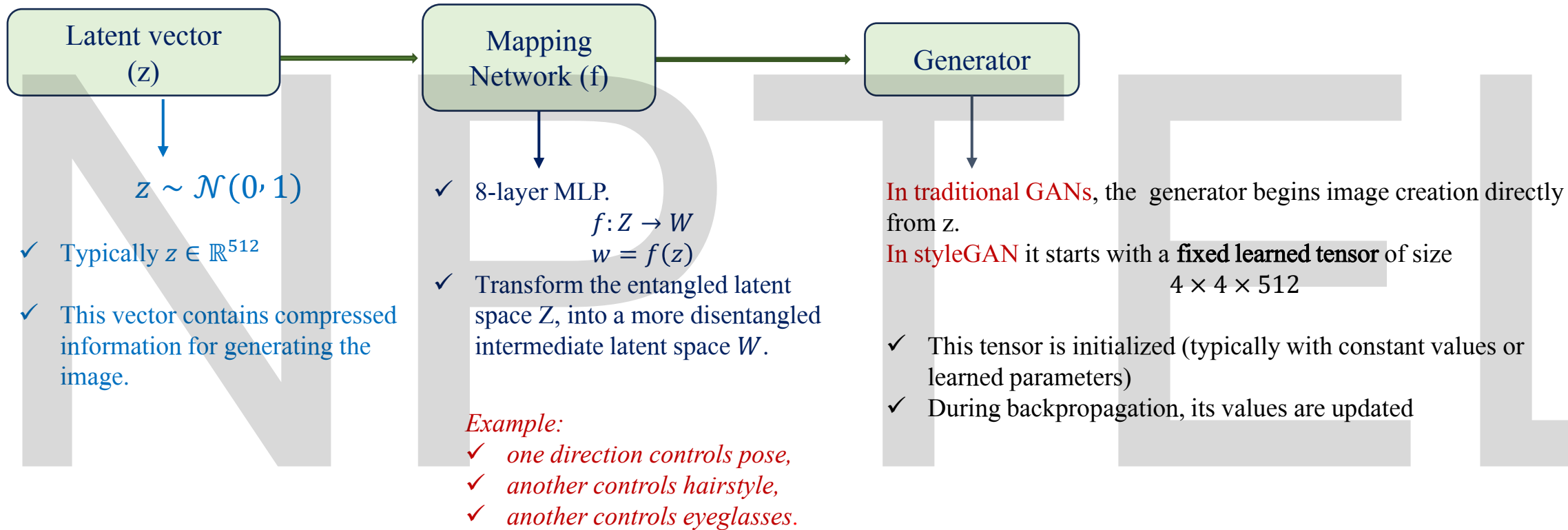


$z \rightarrow \text{Mapping Network} \rightarrow w \rightarrow \text{Styles} \rightarrow \text{Synthesis Network} \rightarrow \text{Image}$

Ref:

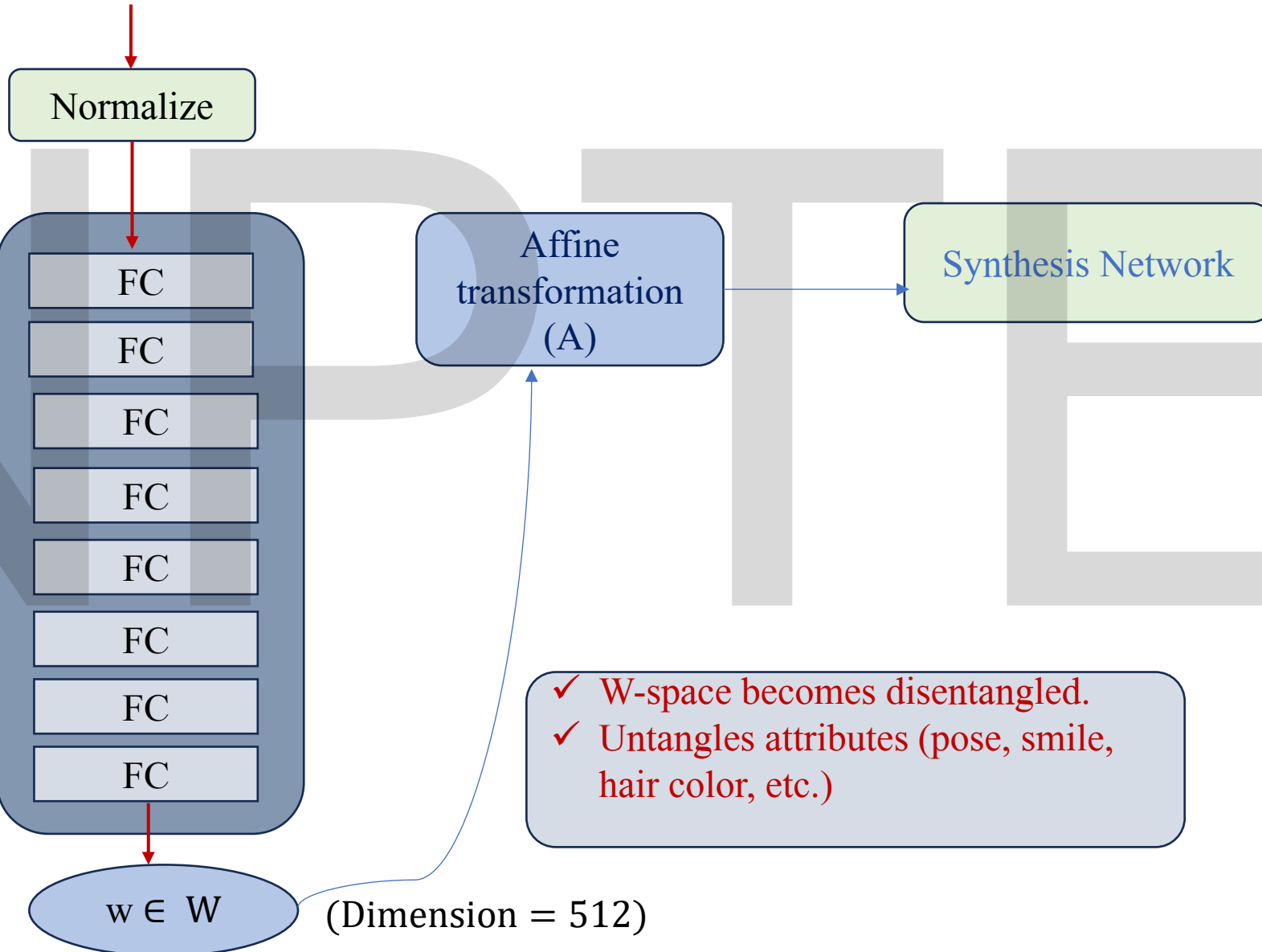
Tero Karras, Samuli Laine, Timo Aila, (Nvidia) "A Style-Based Generator Architecture for Generative Adversarial Networks"

Mapping Network and W Space in StyleGAN



Mapping Network in StyleGAN

Latent $z \in Z$ (Dimension = 512)



Analogy in StyleGAN:



Same starting Clay Block



Final style is different

Styles injected through

- ✓ AdaIN.
- ✓ Noise injection.
- ✓ Not from the initial input tensor.

Affine Transformation in StyleGAN

Affine Transform Produces Styles

$$y = Wx + b$$

In Style GAN:

$$z \rightarrow w \rightarrow y = Aw + b$$

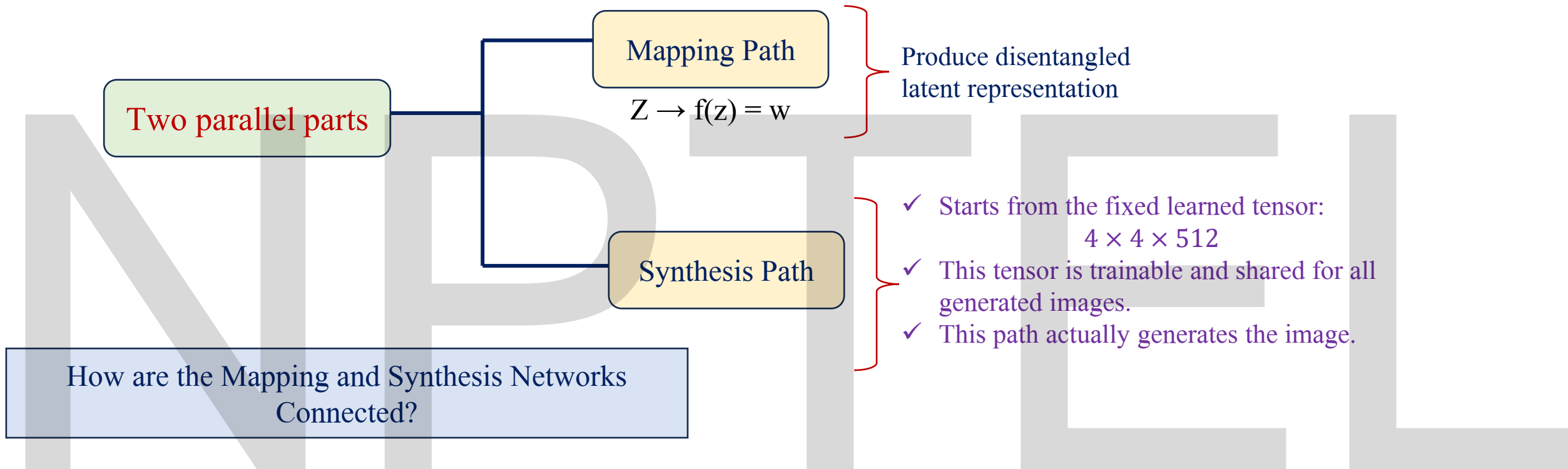
Style Parameters

Injected

AdaIN at every layer

- ✓ The affine transformation converts latent vector w into style parameters y .
- ✓ The Learned tensor provides spatial feature maps (think of it as a lump of clay)
- ✓ Styles = sculpting instructions

Full Architecture Flow of StyleGAN



- The learned tensor flows through convolution layers.(18 conv layers)
 $w \rightarrow \text{Affine Transform} \rightarrow \text{Style parameters}$
- At every convolution layer, the affine transformation produces style parameters that are injected through AdaIN.

Synthesis Network

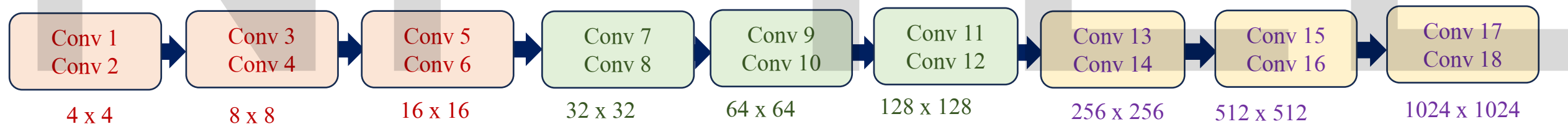


Contains edges, textures, face structure, hair pattern etc

AdaIN

- Normalize the feature map (removes the original statistics of the feature map)
- Apply a new style using learned scale and bias

➤ After 2 Conv layers in one resolution, there is upsampling



Lower resolution learns

- ✓ pose
- ✓ face shape
- ✓ coarse structure

Middle layers learn:

- ✓ eyes
- ✓ nose
- ✓ hairstyle

Higher-resolution layers learn:

- ✓ pores
- ✓ wrinkles
- ✓ hair strands
- ✓ texture

➡ Upsampling

Adaptive Instance Normalization (AdaIN)

Let the feature map be:

$$\text{Let } x_i = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Standard deviation:

$$\text{Mean } \mu(x_i) = \frac{2+4+6+8}{4} = 5$$

$$\sigma(x_i) = \sqrt{\frac{(2-5)^2 + (4-5)^2 + (6-5)^2 + (8-5)^2}{4}}$$

$$= \sqrt{\frac{9 + 1 + 1 + 9}{4}} = \sqrt{5} \approx 2.236$$

Normalize

$$\frac{x_i - \mu(x_i)}{\sigma(x_i)}$$

$$\begin{bmatrix} -1.34 & -0.45 \\ 0.45 & 1.34 \end{bmatrix}$$

- zero mean
- unit variance

- ✓ After normalization old style information is removed
- ✓ Only structural information remains.

AdaIN first normalizes every channel independently

Style Injection in AdaIN

$z \rightarrow \text{Mapping Network} \rightarrow w \rightarrow \text{Affine Transformation}$

$$y = Aw + b$$

$$y = (y_s, y_b)$$

y_s : Learned scaling parameter
 y_b : Learned Bias Parameter

These are used by AdaIN

$$\text{AdaIN}(x_i, y) = y_{s,i} \frac{x_i - \mu(x_i)}{\sigma(x_i)} + y_{b,i}$$

Styled feature maps

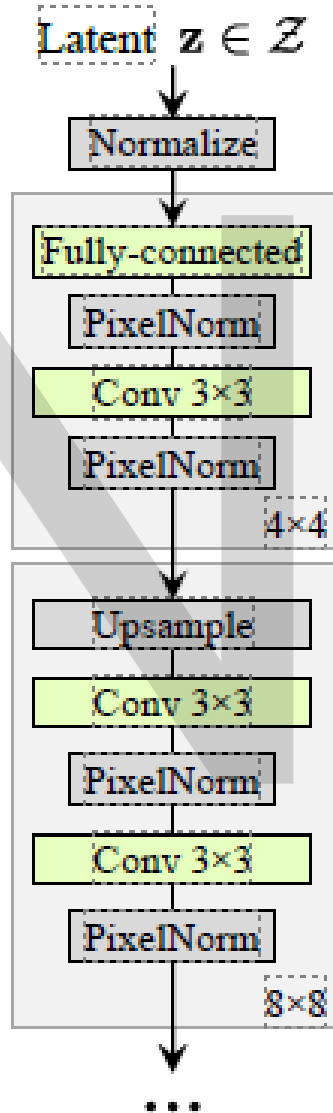
↓
Generated image

If $y_s = 2$ and $y_b = 1$

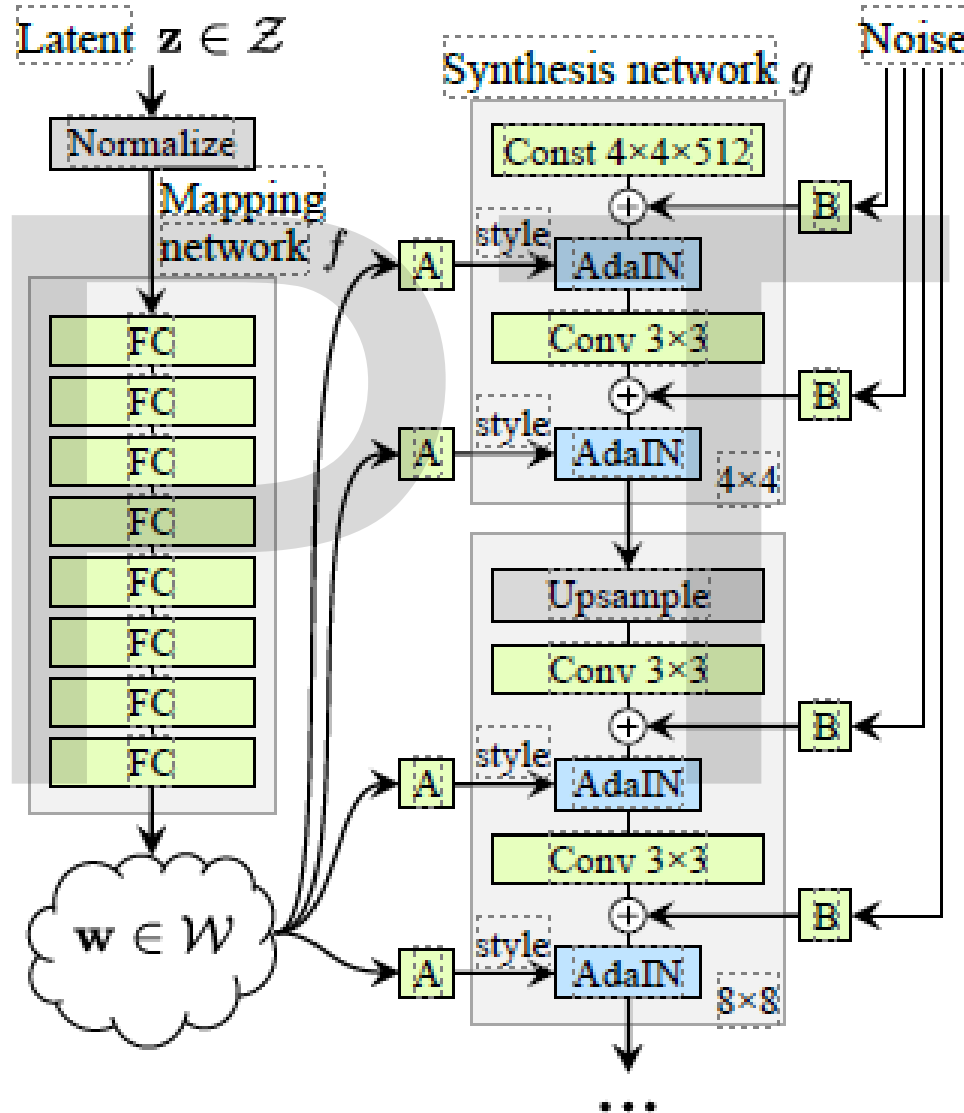
$$\text{AdaIN}(x_i, y) = 2 \times \begin{bmatrix} -1.34 & -0.45 \\ 0.45 & 1.34 \end{bmatrix} + 1$$

$$= \begin{bmatrix} -1.68 & 0.1 \\ 1.9 & 3.68 \end{bmatrix}$$

Style Based Generator



(a) Traditional



(b) Style-based generator

Noise adds stochastic variation to the synthesis network

B:

- Learned scaling factor
- Decides how strongly the noise should affect each feature channel.

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Summary

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Next Session

StyleGAN 2